Study cards, comparisons, and trace explanations

37 catalogued methods | 10,000-evaluation budget where specified | fixed seeds

Executive summary

Performance depends on the problem structure. On P3, MOBA and PESA-II lead with nearly tied hypervolume and low IGD. On P5, many methods reach almost the same HV ceiling, so spacing and IGD are needed to distinguish their fronts. For DTLZ2 with five objectives, HypE has the highest HV while RVEA has lower IGD. On the 15-city P8 TSP, MOACO leads the compared algorithms by mean HV.

How to read the metrics

All objectives are minimised internally. HV uses fixed per-problem normalisation and reference point 1.1; larger is better. IGD measures distance to a reference front and Schott spacing measures regularity; lower is better. Task A3 is one seeded run; Task D aggregates 10 seeds. P5 IGD uses the pooled non-dominated runs, not an analytic global front. Five-objective HV is estimated by Monte Carlo.

Task D representative results

Problem	Top by HV	HV mean +/- SD	IGD mean +/- SD	Interpretation
P3, 2 objectives	MOBA (29)	0.53874 +/- 0.00003	0.00379 +/- 0.00003	PESA-II close second
P5, constrained	SMS-EMOA (33)	1.19580 +/- 0.00000	0.00509 +/- 0.00022	MOBA has lower IGD
DTLZ2, M=5	HypE (34)	1.2324 +/- 0.0066	0.2622 +/- 0.0048	RVEA IGD is 0.1735
P8, 15-city TSP	MOACO (25)	0.7832 +/- 0.0219	0.0108 +/- 0.0182	NSGA-II lowest spacing

A rank by HV is not an all-metric win. On P5, SMS-EMOA ranks first in HV, while MOBA has lower mean spacing and IGD. Values are from taskD_compare.txt; SD is sample SD over 10 seeds.

Task C: trace explanations

Algorithm 8 | TOPSIS on P12

Vector-normalise each criterion, apply weights, and compute distances D+ and D- from the positive and negative ideal. Closeness C = D- / (D+ + D-); larger C ranks better. With weights (0.3, 0.3, 0.2, 0.2), S3 ranks first (0.6298), followed closely by S1 (0.6214), then S5, S2, and S4. Shifting 0.1 weight from criterion 2 to criterion 1 changes the order to S1, S2, S3, S5, S4. The chosen ranking depends on decision-maker preference.

Algorithm 32 | IBEA on the small P1 trace

The additive epsilon indicator compares normalised objective vectors: Ieps(i,j) = max_k(fn_i,k - fn_j,k). IBEA combines pairwise indicators into exponential fitness. At kappa=0.05 the first trace fitness spans -2.6930 to approximately 0; environmental selection repeatedly removes the least-fit member from the parent-offspring union until four remain. Kappa controls the sharpness of the fitness response.
P3 sensitivity (10 seeds): kappa=0.05 gives HV 0.5370 +/- 0.0001 and IGD 0.0085 +/- 0.0004. kappa=0.01 has slightly lower HV (0.5356 +/- 0.0077) but better mean IGD (0.0067 +/- 0.0027). kappa=0.25 is worse on both (HV 0.5165, IGD 0.0453). The preferred setting depends on metric; 0.05 is more stable in HV.

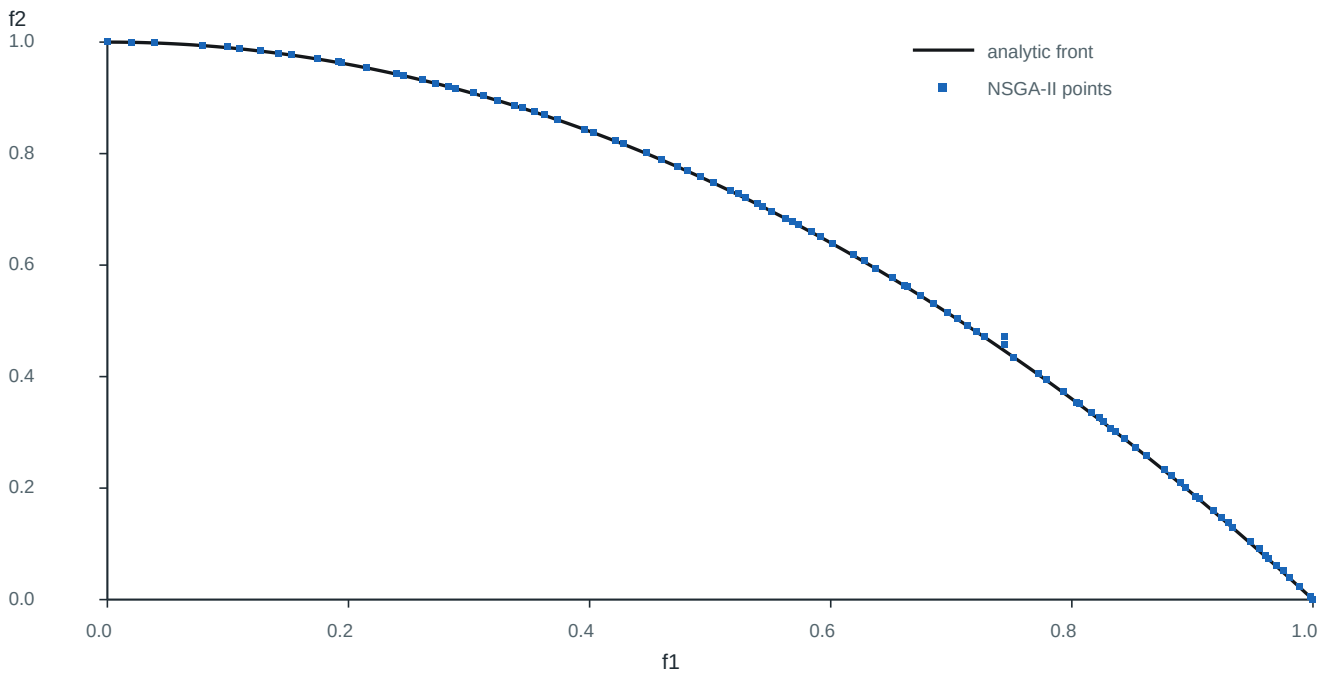
Algorithm 20 | MOEA/D-ACO on the hand-checkable TSP

Each weight-vector subproblem constructs a tour using its pheromone matrix and a distance/time heuristic. A tour replaces neighbouring incumbents when its normalised Tchebycheff value improves; pheromone evaporates by rho and is reinforced along neighbour tours. In iteration 1 the two tours have objectives (15,25) and (13,21); in iteration 2 the reported tours are (13,21) and (13,19). This five-city trace is distinct from the 15-city Task D benchmark.
P8 rho sensitivity: rho=0.02 gives the highest tested mean HV (0.7854 +/- 0.0113); rho=0.10 gives 0.7730 +/- 0.0232; rho=0.50 falls to 0.7203 +/- 0.0523. In this experiment, high evaporation weakens useful pheromone memory.

Source: handtrace_taskC.txt and taskC_sensitivity.txt. Trace values illustrate algorithm steps; they are not the multi-seed benchmark means.

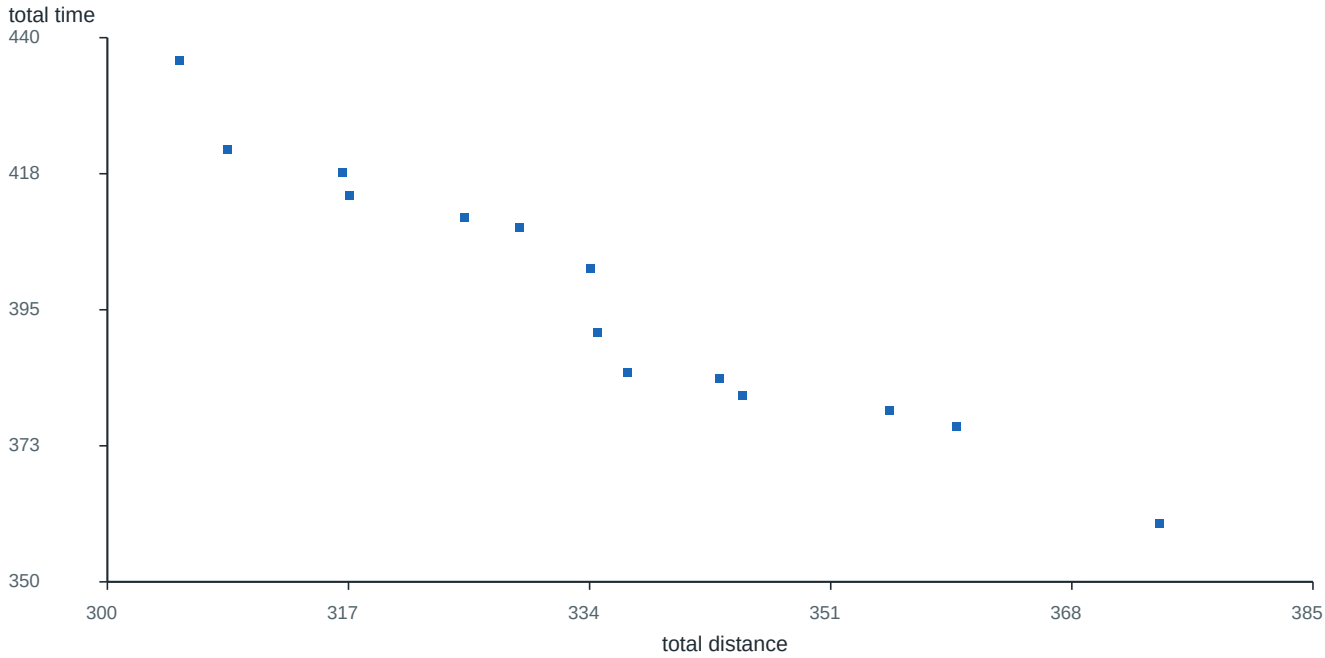
Figures: continuous and discrete trade-offs

Figure 1. NSGA-II on P3 with analytic true front



The points closely follow the analytic curve, consistent with the single-run IGD of 0.00484. The visual overlap makes small density differences difficult to assess.

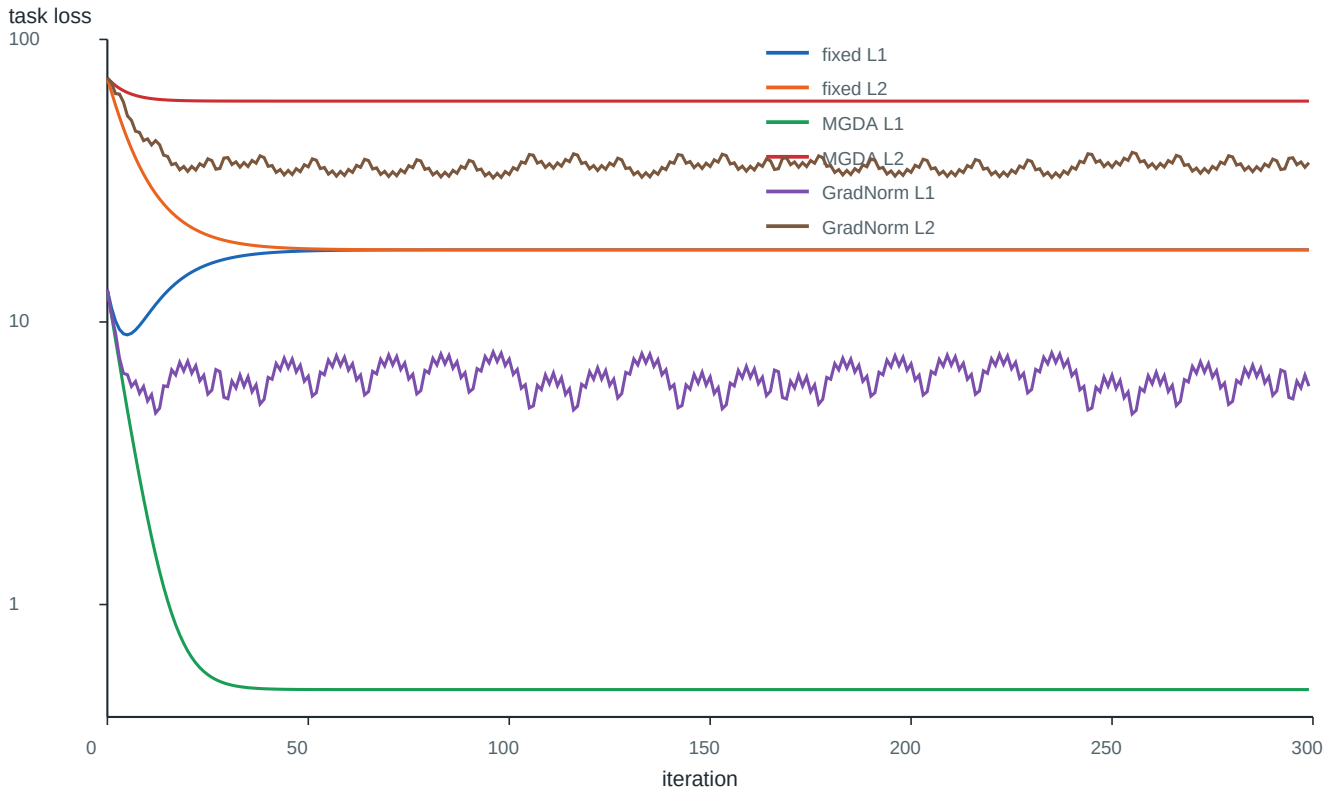
Figure 2. MOEA/D-ACO on the P8 distance-time plane



Lower distance tends to accompany higher travel time. These are single-run tours; no reference front is drawn. Use the multi-seed table for algorithm comparisons.

Figure 3 and study interpretation

P13(a) task losses over 300 iterations (log scale)



Fixed equal weights approach losses (18,18). MGDA sharply lowers L1 while leaving L2 near 60.5, selecting a different Pareto point; GradNorm balances differently and oscillates. The methods encode different trade-offs, not one universally best trajectory.

Other results and limitations

P12: TOPSIS selects S3 while VIKOR at v=0.5 selects S1; weights and v alter rankings. P11: goal programming meets the profit goal but misses the 600-unit target by about 100 units. Lexicographic optimisation selects (300,200), reaching 500 units while preserving profit 2400. P14: the requested 1.8 kg / 12 h target lies behind the front; the best runtime at 1.8 kg is 14.04 h.

Constraint handling matters: A3 plain NSGA-II on P5 returns 53% feasible points, while constrained NSGA-II returns 100% feasible points. A high HV does not certify feasibility.

P9 was not run because its instructor dataset and 5-fold 5-NN evaluation are unavailable; P13(b) is not implemented. P8 is a seeded synthetic 15-city TSP. MOEA/D-ACO, MOBA, MOSA, and MO-TLBO are compact educational variants. Claims are limited by budgets, seeds, and reference-front availability.

Viva prompts: Why can weighted sums miss non-convex fronts? Why can HV and IGD disagree? What changes when feasibility rules are disabled? How do TOPSIS weights and VIKOR v encode preference? How does rho affect pheromone memory?